

Phylogenetic analysis in the predicted secondary structure space

Jhih-Siang Lai, Burkard Rost, Bostjan Kobe and Mikael Bodén

School of Chemistry and Molecular Biosciences, The University of Queensland, Brisbane, QLD 4072, Australia
js.lai@uqconnect.edu.au

Background

The sequence-based 3-class secondary structure prediction reaches reasonable performances. The prediction model can be considered as a mapping from the “sequence” to the “predicted protein fold” space. In the sequence space, a residue has 20 standard amino acid states; in the prediction space, a residue has 60 states of 20 amino-acid states and 3 secondary-structure states jointly. This work proposed a generalised evolutionary model (MODEL60) in the prediction space of PredictProtein [1]. We implemented Needleman-Wunsch's alignment method with the substitution matrix derived from MODEL60. We evaluated alignments with Q and QMEAN [3]. A maximum likelihood tree inference method was implemented with MODEL60. Based on the dataset of Toll/interleukin-1 receptor (TIR) domains proposed in our previous work [4], we compared the phylogenetic trees inferred by MODEL60, amino acid and the DSSP models.

Results

Generalised substitution matrix (amino acid and secondary structure)

EA	EC	ED	EE	EF	EG	EH	EI	EK	EL	EN	EP	EQ	ER	ES	ET	EV	EY	HA	HC	HD	HE	HF	HG	HH	HI	HK	HL	HM	HN	HP	HQ	HR	HS	HT	HV	HW	HY	LA	LC	LD	LE	LF	LG	LH	LI	LK	LL	LN	LP	LQ	LS	LT	LV	LW	LY						
EA	5	17	-6	-2	-4	-3	-1	-5	4	-3	-4	-5	0	2	0	-2	0	-1	13	-8	-6	-3	-5	-5	-4	-7	-5	-5	-7	-3	0	-2	-1	3	2	1	18	-7	-7	-5	-4	-2	-6	-5	-4	-4	-4	-6	-1	1	-2	0	-2	0							
EC	-3	-6	11	-6	-2	3	-9	-3	-11	-9	4	6	-1	4	-2	4	-7	13	-4	-8	4	-11	-3	0	18	-4	-11	18	-2	-2	-7	-3	-5	-8	-14	-6	-2	-6	0	5	-5	-2	1	-8	-2	-10	-8	0	6	11	5	-2	3	-6	-13	-4					
ED	-1	-4	0	11	-3	-2	-7	0	-9	-6	1	5	8	-3	-3	-24	-2	-2	0	-5	4	-9	-6	-2	7	2	5	0	6	-5	-7	3	0	2	8	-7	2	4	-9	7	2	0	18	-6	-14	-16	-1	7	0	11	-5	-3	0	-9	-4						
EE	-2	-5	-13	0	13	-3	-2	-10	-1	0	1	9	3	-7	-4	-22	-8	-2	1	13	13	0	12	-2	8	-3	-18	-18	-11	-5	-7	-4	-8	-2	10	-12	-12	-5	-13	6	-4	1	9	-9	-18	2	0	4	-6	-12	-9										
EF	-1	-4	-2	-3	-13	1	-2	-10	-6	-11	-9	3	-7	-6	-4	-1	-5	-7	-9	-11	-2	-5	-3	-4	13	0	-6	-9	-7	-11	-8	-3	-7	-7	-5	-1	-7	-10	-12	-1	-3	-2	-12	18	-5	-9	-5	-10	-8	2	6	-5	3	0	-4	-6	-8	-10			
EG	-5	-3	0	-1	-2	-3	10	0	18	-6	-7	3	4	0	3	-54	-8	-6	-5	-51	-4	-3	-5	18	-9	-2	-8	-8	-1	-5	2	0	4	-5	-9	-8	-3	-5	3	0	2	-2	-5	16	-1	-6	-7	2	4	3	1	-3	-4	-7	-7	4					
EH	-2	-5	-10	-4	0	1	-2	-6	-4	-7	0	4	4	0	0	18	-3	-18	-9	-3	-5	-6	3	0	18	-4	-7	-4	-1	-11	-9	2	3	-9	-5	-16	-5	-1	-5	-16	-5	0	-6	-3	-1	15	3	-1	-5	-8	-8										
EI	-4	-5	-3	1	-10	-6	0	-5	11	0	6	-5	1	4	6	-2	1	-5	-9	-5	-7	-4	1	12	-6	-2	-7	-8	-6	0	6	2	4	-3	-7	-10	-10	-3	-5	0	-18	-5	-1	-5	0	-6	3	-3	-1	-5	-8	-8									
EK	-4	-5	-3	1	-10	-6	0	-5	11	0	6	-5	1	4	6	-2	1	-5	-9	-5	-7	-4	1	12	-6	-2	-7	-8	-6	0	6	2	4	-3	-7	-10	-10	-3	-5	0	-18	-5	-1	-5	0	-6	3	-3	-1	-5	-8	-8									
EL	-5	-4	-11	-9	-1	-11	-6	3	-6	9	5	-7	3	-5	-4	-3	2	-6	-4	-5	-5	-11	18	0	18	-7	-2	0	6	3	-8	-4	-5	-8	-4	-3	0	-7	-5	-5	-11	-9	-1	-11	-6	-3	-7	0	-3	-8	-4	-4	-6	-5							
EN	-3	-3	-9	-4	0	1	-9	-3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0								
EP	-1	-3	4	-1	-2	-5	-3	-5	2	2	5	1	4	0	1	3	1	-5	-11	-3	-3	-2	3	-8	-3	-2	-7	0	-8	-6	0	5	-5	-2	3	1	-6	-12	-4	-1	-3	0	-7	-5	-4	3	-6	-4	1	-1	3	1	-1	-4	-11	-9					
EQ	-3	-5	-1	-1	-6	-4	-3	-3	0	-2	18	3	-2	-2	-5	-8	-6	-4	-7	-2	1	-9	-6	-2	7	2	-5	-5	-3	0	1	-3	-6	-6	-9	-7	-3	-5	-1	2	-6	-3	0	-3	-3	-1	2	0	-2	-5	-7	-4									
ER	-4	-6	-4	-3	-6	-4	-1	-1	0	2	2	3	-6	-2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0								
ES	4	-2	-3	-3	-1	-3	-4	-2	-4	-3	3	1	-2	-2	0	4	-3	-6	-4	2	1	-3	-4	-5	-1	-4	-4	-3	1	-1	-3	-4	6	2	-3	-9	-5	3	-2	-3	-4	-1	-3	-3	-3	2	0	3	-3	-7	3	-2	-8	-4							
ET	3	0	-4	-3	-7	-5	-4	0	-1	-3	0	1	-1	-2	-3	4	0	0	11	-7	1	-2	-4	-4	-7	-5																																			
EV	-1	-2	-3	-5	-11	-10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0								
EY	-10	-2	-13	-12	-2	-9	-6	-10	-9	-6	-7	-11	-12	-8	-3	-8	-11	-10	1	1	1	-2	-14	-14	-9	-8	-11	-10	-7	-9	-12	-12	-9	-2	-4	-8	-11	-11	1	-1	-1	-2	-14	-13	-5	-9	-2	-10	-5	-3	-12	-11	-7	3	-8	-11	-10				
HA	-8	-6	-4	-6	-1	-1	-4	-8	-9	-4	-6	-3	-10	-6	-6	-4	-7	-8	1	10	-2	-7	-11	5	12	3	-8	-10	-5	-2	-4	-1	-1	-7	-9	-6	-9	-9	-2	11	-8	0	-6	-9	-7	-12	3	-7	-9	-3	-7	-4	-10	-6	-7	-5	-8	-8	-1		
HC	5	1	-2	-4	-3	-5	-3	-3	-2	-4	-6	2	3	1	-1	-1	-1	1	2	2	1	2	2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
HD	-3	-6	11	-6	-2	3	-9	-3	-11	-9	4	6	-1	4	-2	4	-7	13	-4	-8	4	-11	-3	0	18	-4	-11	18	-2	-2	-7	-3	-5	-8	-14	-6	-2	-6	0	5	-5	-2	1	-8	-2	-10	-8	0	6	11	5	-2	3	-6	-13	-4					
HE	-1	-4	0	11	-3	-2	-7	0	-9	-6	1	5	8	-3	-3	-24	-2	-2	0	-5	4	-9	-6	-2	7	2	5	0	6	-5	-7	3	0	2	8	-7	2	4	-9	7	2	0	18	-6	-14	-16	-1	7	0	11	-5	-3	0	-9	-4						
HF	-2	-5	-13	0	13	-3	-2	-10	-1	0	1	9	3	-7	-4	-22	-8	-2	1	13	13	0	12	-2	8	-3	-18	-18	-11	-5	-7	-4	-8	-2	10	-12	-12	-5	-13	6	-4	1	9	-9	-18	2	0	4	-6	-12	-9										
HG	-1	-4	-2	-3	-13	1	-2	-10	-6	-11	-9	3	-7	-6	-4	-1	-5	-7	-9	-11	-2	-5	-3	-4	13	0	-6	-9	-7	-11	-8	-3	-7	-7	-5	-1	-7	-10	-12	-1	-3	-2	-12	18	-5	-9	-5	-10	-8	2	6	-5	3	0	-4	-6	-8	-10			
HH	-5	-3	0	-1	-2	-3	10	0	18	-6	-7	3	4	0	3	-54	-8	-6	-5	-51	-4	-3	-5	18	-9	-2	-8	-8	-1	-5	2	0	4	-5	-9	-8	-3	-5	3	0	2	-2	-5	16	-1	-6	-7	2	4	3	1	-3	-4	-7	-7	4					
HI	-2	-5	-10	-4	0	1	-2	-6	-4	-7	0	4	4	0	0	18	-3	-18	-9	-3	-5	-6	3	0	18	-4	-7	-4	-1	-11	-9	2	3	-9	-5	-16	-5	-1	-5	-16	-5	0	-6	-3	-1	15	3	-1	-5	-8	-8										
HL	-4	-5	-3	1	-10	-6	0	-5	11	0	6	-5	1	4	6	-2	1	-5	-9	-5	-7	-4	1	12	-6	-2	-7	-8	-6	0	6	2	4	-3	-7	-10	-10	-3	-5	0	-18	-5	-1	-5	0	-6	3	-3	-1	-5	-8	-8									
HM	-5	-4	-11	-9	-1	-11	-6	3	-6	9	5	-7	3	-5	-4	-3	2	-6	-4	-5	-5	-11	18	0	18	-7	-2	0	6	3	-8	-4	-5	-8	-4	-3	0	-7	-5	-5	-11	-9	-1	-11	-6	-3	-7	0	-3	-8	-4	-4	-6	-5							
HN	-3	-3	-9	-4	0	1	-9	-3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0								
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HQ	-3	-5	-1	-1	-6	-4	-3	-3	0	-2	18	3	-2	-2	-5	-8	-6	-4	-7	-2	1	-9	-6	-2	7	2	-5	-5	-3	0	1	-3	-6	-6	-9	-7	-3	-5	-1	2	-6	-3	0	-3	-3	-1	2	0	-2	-5	-7	-4									
HR	-4	-6	-4	-3	-6	-4	-1	-1	0	2	2	3	-6	-2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0								
HS	4	-2	-3	-3	-1	-3	-4	-2	-4	-3	3	1	-2	-2	0	4	-3	-6	-4	2	1	-3	-4	-5	-1	-4	-4	-3	1	-1	-3	-4	6	2	-3	-9	-5	3	-2	-3	-4	-1	-3	-3	-3	2	0	3	-3	-7	3	-2	-8	-4							
HT	3	0	-4	-3	-7	-5	-4	0	-1	-3	0	1	-1	-2	-3	4	0	0	11	-7	1	-2	-4	-4	-7	-5																																			
HV	-1	-2	-3	-5	-11	-10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0							
HW	-10																																																												